

A Counterexample to a Resolvent-to-Semigroup Eventual Positivity Question

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Source. Daniel Daners, Jochen Glück, and James B. Kennedy, *Eventually and asymptotically positive semigroups on Banach lattices*, arXiv:1511.05294v2.

Target question. The relevant source question is Open Problem (d) in Section 10. In the notation of the source, it asks whether a C_0 -semigroup whose generator has dominant spectral value $s(A)$ must be individually eventually positive if the resolvent is individually eventually positive at $s(A)$, perhaps after adding regularity assumptions.

Suppose that all assumptions of Theorem 5.2 are fulfilled except for the condition $D(A) \subseteq E_u$. Can individual eventual strong positivity of the semigroup still be characterised by individual eventual strong positivity of the resolvent at $s(A)$ plus a spectral condition?

(c) We only defined the notion of asymptotic positivity of a semigroup $(e^{tA})_{t \geq 0}$ under the assumption that the rescaled semigroup $(e^{t(A-s(A))})_{t \geq 0}$ be bounded. If this assumption is not fulfilled, it is not clear to the authors if the condition $d_+(e^{t(A-s(A))}f) \rightarrow 0$ for each $f \geq 0$ should still be used to define individual asymptotic positivity, or if for instance the condition $\frac{d_+(e^{t(A-s(A))}f)}{\|e^{t(A-s(A))}\|} \rightarrow 0$ for each $f \geq 0$ would be more appropriate. The same question arises for asymptotic positivity of the resolvent if the Abel-boundedness condition in Definition 7.2 is dropped:

How should asymptotic positivity of semigroups and resolvents be defined without additional boundedness assumptions?

(d) The following problem is concerned with eventual positivity rather than eventual strong positivity: In Example 7.1(b) and in [18, Example 8.2] we showed that individual eventual positivity of a semigroup does not imply individual eventual positivity of the resolvent at $s(A)$, even in finite dimensions. However, if the spectral bound $s(A)$ is a dominant spectral value, one might ask whether at least the converse implication is true:

Let $(e^{tA})_{t \geq 0}$ be a C_0 -semigroup with dominant spectral value $s(A)$ of A and suppose that the resolvent is individually eventually positive at $s(A)$. Does it follow (maybe under some additional regularity assumptions) that $(e^{tA})_{t \geq 0}$ is individually eventually positive?

Answer. The implication is false in its base form, even for a bounded matrix on finite-dimensional Euclidean space with the standard cone. Thus finite dimensionality, boundedness, uniform continuity, and analyticity are not enough as additional regularity assumptions.

Definitions Used

On $\mathbb{C}^n$ we use the standard Banach lattice structure: a vector is positive if all of its coordinates are real and non-negative. A semigroup $(e^{tA})_{t \geq 0}$ is individually eventually positive if for every $0 \leq f$ there is $t_f \geq 0$ such that $e^{tA}f \geq 0$ for all $t \geq t_f$.

For a real spectral value λ_0 , the resolvent is individually eventually positive at λ_0 if, for every $0 \leq f$, $R(\lambda, A)f \geq 0$ for all real $\lambda > \lambda_0$ sufficiently close to λ_0 . The example below satisfies a stronger property: $R(\lambda, A)$ is a positive operator for every $0 < \lambda \leq 1$.

Proof Intuition

The construction separates the Laplace-average behaviour of the resolvent from the pointwise-in-time behaviour of the semigroup. A stable three-dimensional block is chosen to be e^{-t} times a rotation around the positive vector $(1, 1, 1)$. The semigroup therefore keeps acquiring negative diagonal entries at arbitrarily large times, although their size decays. The resolvent, however, integrates this oscillation against $e^{-\lambda t}$; choosing the rotation frequency large enough makes this averaged operator positive near $\lambda = 0$. Adding one independent zero eigenvalue makes the spectral bound equal to the dominant spectral value 0 without changing the sign-changing stable block.

Theorem 1. *There is a bounded real operator A on the Banach lattice $\mathbb{C}^4$ such that:*

1. $s(A) = 0$ is a dominant spectral value of A ;
2. $R(\lambda, A)$ is positive for every $0 < \lambda \leq 1$;
3. the semigroup $(e^{tA})_{t \geq 0}$ is not individually eventually positive.

Proof. Let

$$K = \frac{1}{\sqrt{3}} \begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix} \quad \text{on } \mathbb{R}^3.$$

Let $P = \frac{1}{3}\mathbf{1}\mathbf{1}^T$, where $\mathbf{1} = (1, 1, 1)^T$, and let $H = I - P$. Then P is the projection onto $\text{span}\{\mathbf{1}\}$, and direct calculation gives

$$KP = PK = 0, \quad K^2 = -H.$$

In particular, K has eigenvalues $0, i, -i$.

Set

$$B = -I_3 + 4K \quad \text{and} \quad A = 0 \oplus B$$

on $\mathbb{R} \oplus \mathbb{R}^3$ and then complexify to $\mathbb{C}^4$. Since the spectrum of B is $\{-1, -1 + 4i, -1 - 4i\}$, we have

$$\sigma(A) = \{0, -1, -1 + 4i, -1 - 4i\}.$$

Thus $s(A) = 0$ and the peripheral spectrum is just $\{0\}$, so 0 is a dominant spectral value.

We next prove positivity of the resolvent near 0. Let $0 < \lambda \leq 1$ and write $r = 1 + \lambda$, so $1 < r \leq 2$. Since $K^2 = -H$ and $PK = 0$, we have

$$R(\lambda, B) = (rI - 4K)^{-1} = \frac{1}{r}P + \frac{r}{r^2 + 16}H + \frac{4}{r^2 + 16}K.$$

The diagonal entries of this matrix are

$$\frac{1}{3r} + \frac{2r}{3(r^2 + 16)} > 0.$$

For an off-diagonal entry where the corresponding entry of K is $+1/\sqrt{3}$, the entry is

$$\frac{1}{3r} - \frac{r}{3(r^2 + 16)} + \frac{4}{\sqrt{3}(r^2 + 16)} > 0.$$

For an off-diagonal entry where the corresponding entry of K is $-1/\sqrt{3}$, the entry is

$$\frac{1}{3r} - \frac{r}{3(r^2 + 16)} - \frac{4}{\sqrt{3}(r^2 + 16)}.$$

Multiplying this last expression by the positive number $3r(r^2 + 16)$ gives

$$16 - 4\sqrt{3}r \geq 16 - 8\sqrt{3} > 0.$$

Hence every entry of $R(\lambda, B)$ is positive for $0 < \lambda \leq 1$. Consequently

$$R(\lambda, A) = \text{diag}(1/\lambda, R(\lambda, B))$$

is a positive operator for every $0 < \lambda \leq 1$. This is stronger than individual eventual positivity of the resolvent at $s(A) = 0$.

It remains to show that the semigroup is not individually eventually positive. The same identities for K give

$$e^{tB} = e^{-t}e^{4tK} = e^{-t}(P + \cos(4t)H + \sin(4t)K).$$

Let $e_1 = (1, 0, 0)^T$. For

$$t_n = \frac{(2n+1)\pi}{4} \quad (n \in \mathbb{N})$$

we have $\cos(4t_n) = -1$ and $\sin(4t_n) = 0$. Therefore the first coordinate of $e^{t_n B}e_1$ is

$$e^{-t_n} \left(\frac{1}{3} - \frac{2}{3} \right) = -\frac{1}{3}e^{-t_n} < 0.$$

The positive vector $f = (0, e_1) \in \mathbb{R} \oplus \mathbb{R}^3$ is thus mapped outside the positive cone at the arbitrarily large times t_n . Hence there is no t_f after which $e^{tA}f$ is always positive, and the semigroup is not individually eventually positive. $\square$

Verification Notes

The proof above is exact. The script `code/verify_counterexample.py` performs a numerical sanity check of the displayed identities: it samples $\lambda \in (0, 1]$ to confirm positivity of the resolvent matrix and samples the times $t_n = (2n+1)\pi/4$ to confirm the negative first coordinate of $e^{t_n B}e_1$.

The bounded literature check used the run indexes for arXiv:1511.05294 and keywords including “individual eventual positivity”, “resolvent”, “dominant spectral value”, and “eventually positive semigroup”. It also checked locally cached later papers citing the source, including arXiv:2210.06135 and arXiv:2405.16371, and web queries for the exact resolvent-to-semigroup phrasing. No existing answer to Open Problem (d) was found in those checks.

Scope

This disproves the unqualified implication in Open Problem (d), and it rules out several natural regularity repairs such as finite dimensionality, boundedness of the generator, uniform continuity, and analyticity. It does not preclude a positive theorem under a different additional hypothesis designed to exclude the stable oscillatory block used here.

References

- [1] Daniel Daners, Jochen Glück, and James B. Kennedy, *Eventually and asymptotically positive semigroups on Banach lattices*, J. Differential Equations 261 (2016), 2607–2649; arXiv:1511.05294.